PRIVACY AND SECURITY IN CRITICAL INFRASTRUCTURE

PWS as a Mass Surveillance Primitive

RESEARCH FOCUS

3GPP TS 38.331 Protocol Analysis

5G NR Public Warning System (PWS)

Presenters: Dias Amitullayev, Nandana Harikumar

What We Cover Today

01 **Motivation**

Attacks on 5G, why it matters

02 **The Known Threat**

Prior work: Bitsikas & Pöpper (2022)

03 **The Threat Model**

On-demand SIB delivery as a surveillance primitive

04 **RQ1 - Protocol Proof**

3GPP spec + UERANSIM testbed evidence

05 **RQ2 - Physical Proof**

Monte Carlo simulation + real-world validation

06 **Limitations & Fixes**

Limitations and countermeasures

Motivation

- Paging occasions are deterministic - computed from IMSI (International Mobile Subscriber Identity) , identical in 4G and 5G.
- An attacker with a $200 sniffer can observe paging patterns and verify a target's presence in a cell area with fewer than 10 calls.
- No authentication exists at the paging broadcast layer - the UE trusts all paging messages unconditionally.
- The root cause is structural: paging synchronization between UE and network leaks timing side-channel information.

- Hussain et al. exploited paging to identify individual devices (targeted surveillance - needs victim's phone number, e-mail address or social network handler). Also, IMSI was disclosed partially. (7 trailing IMSI bits. The remaining could be bruteforced).
- Lee et al. showed PWS alerts can be spoofed in LTE - and showed that it could lead misinformation/DoS.
- Bitsikas & Pöpper extended the state of the art to PWS spoofing and PWS suppression in 5G.

Key takeaway: passive observation of paging behaviour reveals private information without any cryptographic capability.

How PWS Works

BACKGROUND - PUBLIC WARNING SYSTEM

1. **Public Authority (CBE – Central Broadcast Entity)**
   Government agency detects emergency

2. **Core Network (Cell Broadcast Center Function → Authentication and Mobility Function)**
   Alert propagates through 5G core

3. **Base Station (gNodeB)**
   Sends paging with Paging Radio Network Temporary Identifier (P-RNTI)

4. **All UEs receive System Information Blocks (SIB) 6/7/8 (ETWS Primary, ETWS Secondary, remaining alerts)**
   Phones display alert - no verification

No Authentication

UEs accept PWS from ANY serving cell.
No cryptographic check.

Mandatory Support

All 3GPP devices MUST support PWS.
Cannot be disabled by users.

Broadcast Trigger

P-RNTI reaches ALL devices
in the cell simultaneously.

How PWS Works

- A public authority sends a warning to a Cell Broadcast Entity.
- This propagates through the core network to base stations (gNodeBs in 5G).
- The gNodeB sends a paging message to all UEs in the cell with cause "Emergency".

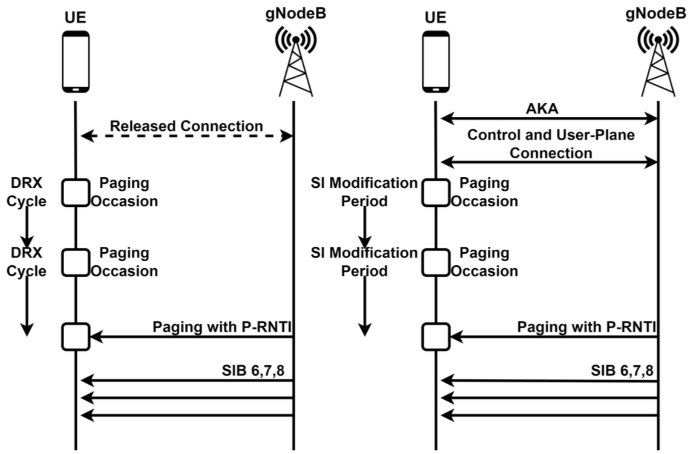


Figure 2: Warning procedure when the UE is RRC-Idle or Inactive (Left) and when in RRC-Connected state (Right)

- UEs in **RRC-Idle or RRC-Inactive state** wake up on their paging occasion and receive the SIB (System Information Block Message) carrying the alert.
- UEs in RRC-Connected state receive it via the System Information modification period.
- The UE displays the alert. There is no verification of source, no cryptographic check, no authentication.

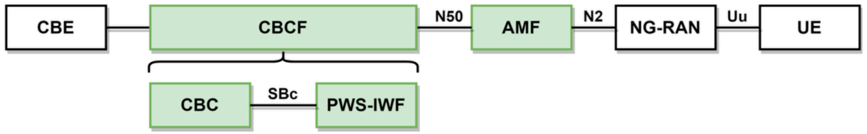


Figure 1: 5G PWS Architecture

Cell Broadcast Entity -> Cell Broadcast Center/ Function -> Access and Mobility Management Function -> NG-RAN -> User Equipment.

The Known Threat: Lack of cryptographic implementations

PRIOR WORK

2019 / 2021

Lee et al.

LTE Presidential Alert Spoofing

- Commodity SDR (Software Defined Radio) hardware (~$200)
- 90% success rate across 49,300 seats
- Four 1-Watt base stations are sufficient
- Proved: no authentication in 4G LTE

2022

Bitsikas & Pöpper

5G Standalone PWS Spoofing

- Extended to 5G networks
- Validated on 5 commercial devices
- Proved flaw is in 3GPP protocol, not hardware
- Broadcast delivery leads to DoS / panic / spread of false information

Sources:[3], [4]

Our Takeaway from the Papers

- **Paging is deterministic and unauthenticated** - wake-up schedule derived from the IMSI, fixed by design, identical in 4G and 5G; the UE trusts every paging message with no cryptographic check.
- **Targeted tracking is already proven** - ToRPEDO attack confirms a victim's presence in a cell with a sniffer and fewer than 10 calls - but requires knowing the victim's personal information (i.e phone number).
- **PWS warning messages can be spoofed** - Lee et al. and Bitsikas et al. both validated fake-alert injection on real commercial devices using commodity hardware.
- **The flaw is in the protocol, not the hardware** - intentional 3GPP design affects every compliant network and device, not just one vendor.
- **All prior work assumes broadcast, one-way delivery** - the network talks, the phone only listens; the worst case studied is misinformation / panic / DoS.

PWS spoofing is practical, cheap and works on real deployed networks today.

Presenter: Dias

Sources:

Our hypothesis: A spoofed PWS alert acts as a Bulk Surveillance Primitive, forcing all "silent" devices to simultaneously reveal their presence

Aspect	Bitsikas (2022)	Our Work
Delivery Mode	Broadcast	On-Demand
Attack Goal	Fake alerts / DoS	Population counting
Uplink Leak	None - passive receive	PRACH burst - detectable
Privacy Risk	Misinformation	Zero-knowledge surveillance

The Attack Equation

PWS Spoofing

Known - Bitsikas 2022

+ On-Demand SIB

Config via attacker SIB1

= PRACH Burst

All phones respond

= Surveillance

Count population - zero-knowledge

How the attack works

Rogue Base Station

Broadcast MIB and SIB1

SIB has si-BroadcastStatus = notBroadcasting

Broadcasted Spoofed ETWS Paging

User Entity

UE checks SIB1: SIB6/7/8 set to notBroadcasting
UE must request via Random Access

Rogue Base Station

Prach Preamble (msg 1)

Random Access Respone (msg 2)

Setup Request (msg 3)

RRC Setup (msg 4)

RRC Connected

User Entity

Attacker

Passively Eavesdropping

3GPP TS 38.331 version 17.13.0 Release 17

5.2.2.3.1 Acquisition of MIB and SIB1

The UE shall:

1> apply the specified BCCH configuration defined in 9.1.1.1;

1> if the UE is in RRC_IDLE or in RRC_INACTIVE; or

1> if the UE is in RRC_CONNECTED while T311 is running:

2> acquire the *MIB*, which is scheduled as specified in TS 38.213 [13];

5.2.2.3.3 Request for on demand system information

2> trigger the lower layer to initiate the Random Access procedure on normal uplink in accordance with TS 38.321 [3] using the PRACH preamble(s) and PRACH resource(s) in *si-RequestConfigRedcap* corresponding to the SI message(s) that the UE requires to operate within the cell, and for which *si-BroadcastStatus* is set to *notBroadcasting*;

5.1.2 Random Access Resource selection

1> if the Random Access procedure was initiated for SI request (as specified in TS 38.331 [5]); and

1> if ra-AssociationPeriodIndex and si-RequestPeriod are configured:

2> determine the next available PRACH occasion from the PRACH occasions corresponding to the selected SSB in the association period given by ra-AssociationPeriodIndex in the si-RequestPeriod permitted by the restrictions given by the ra-ssb-OccasionMaskIndex if configured

01

Clause 5.2.2.3.1

Page No. 65 Source Link

If the UE is in RRC_IDLE or in RRC_INACTIVE … acquire the SIB1 which is scheduled as specified in TS 38.213.

02

Clause 5.2.2.3.3

Page No. 68 Source Link

Set the requested-SI-List to indicate the SI message(s) … for which si-BroadcastStatus is set to notBroadcasting

03

Clause 5.1.2

Page No. 30 Source Link

If the Random Access procedure was initiated for SI request (as specified in TS 38.331) …

GAP: UEs have NO mechanism to authenticate SIB1 or verify cell legitimacy at the RRC layer

Why PWS Specifically Enables This

Mandatory Support

All 3GPP-compliant devices **MUST** support PWS (TS 22.268). Cannot be disabled. Universal attack surface across billions of devices.

Broadcast Trigger

P-RNTI = 0xFFFE reaches **ALL** devices. Single 1-packet message creates synchronized N-device response.

No Authentication

No cryptographic verification of SIB1 content or cell identity at RRC layer.

On-Demand Permitted

3GPP TS 38.331 explicitly allows si-BroadcastStatus = notBroadcasting for SIB6/7/8, converting passive receivers into active transmitters

Difference from Paging

The unique combination: mandatory support + broadcast reach + no authentication + on-demand forcing PRACH + all happening simultaneously

Presenter: Nandana Harikumar

Sources: https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=639

RQ1: PROTOCOL VALIDATION

RESEARCH QUESTION 1 - CAN PWS FORCE A DETECTABLE UPLINK?

Presenter: Nandana Harikumar

UERANSIM Testbed Configuration

1. TESTBED STACK

Component	Version	Role
UERANSIM	v3.2.8	5G NR UE + gNB simulator (3GPP Rel.15/16)
Open5GS	v2.7.7	5G Core (AMF, SMF, UPF + 8 NFs)
MongoDB	6.0	Subscriber database
Platform	Ubuntu 24.04	GitHub Codespaces

2. SOURCE MODIFICATION

File	Change
cli_cmd.hpp	Added PWS_ALERT to GnbCliCommand enum
cli_cmd.cpp	Added pws-alert CLI command parser
cmd_handler.cpp	Added PWS_ALERT case handler
task.hpp	Declared sendPwsEtwsPaging() function
handler.cpp (gNB)	Implemented ETWS paging with ETWS indicator
handler.cpp (UE)	Added ETWS paging detection + PRACH trigger

```
struct GnbCliCommand
{
    enum PR
    {
        PWS_ALERT,
        STATUS,
```

Cli_cmd.hpp

```
namespace nr::gnb
class GnbRrcTask : public NtsTask
    // PWS Attack
    void sendPwsEtwsPaging();
};
```

task.hpp

```
{"ue-release", {"Request a UE context r
{"pws-alert", {"Trigger PWS ETWS alert"
```

Cli_cmd.cpp

```
void GnbRrcTask::sendPwsEtwsPaging()
{
    m_logger->info("[PWS-ATTACK] Broadcasting ETWS paging message (P-RNTI broa
    m_logger->info("[PWS-ATTACK] SIB1 configured with notBroadcasting for SIB6
    m_logger->info("[PWS-ATTACK] UEs in RRC_IDLE must request SIB6 via Random

    paging->lateNonCriticalExtension = asn::New<OCTET_STRING_t>();
    uint8_t etwsFlag = 0x02;
    OCTET_STRING_fromBuf(paging->lateNonCriticalExtension,
                         (const char *)&etwsFlag, 1);

    m_logger->info("[PWS-ATTACK] Sending broadcast paging — all idle UEs will

    // Broadcast to all UEs
    sendRrcMessage(pdu);
    asn::Free(asn_DEF_ASN_RRC_PCCH_Message, pdu);
```

handler.cpp (Node)

```
    }
    case app::GnbCliCommand::PWS_ALERT: {
        m_base->rrcTask->sendPwsEtwsPaging();
        sendResult(msg.address, "PWS ETWS alert triggered");
        break;
    }
```

cmd_handler.cpp

```
void UeRrcTask::receivePaging(const ASN_RRC_Paging &msg)
{
    // === PWS ATTACK DETECTION ===
    if (msg.lateNonCriticalExtension != nullptr &&
        msg.lateNonCriticalExtension->size > 0 &&
        (msg.lateNonCriticalExtension->buf[0] & 0x02))
    {
```

handler.cpp (UE)

Presenter: Nandana Harikumar

Disclosure: *AI was used to assist with code modifications*

Sources: https://github.com/aligungr/UERANSIM , [4]

3. EXPERIMENT SETUP

1. Rogue gNB Configuration

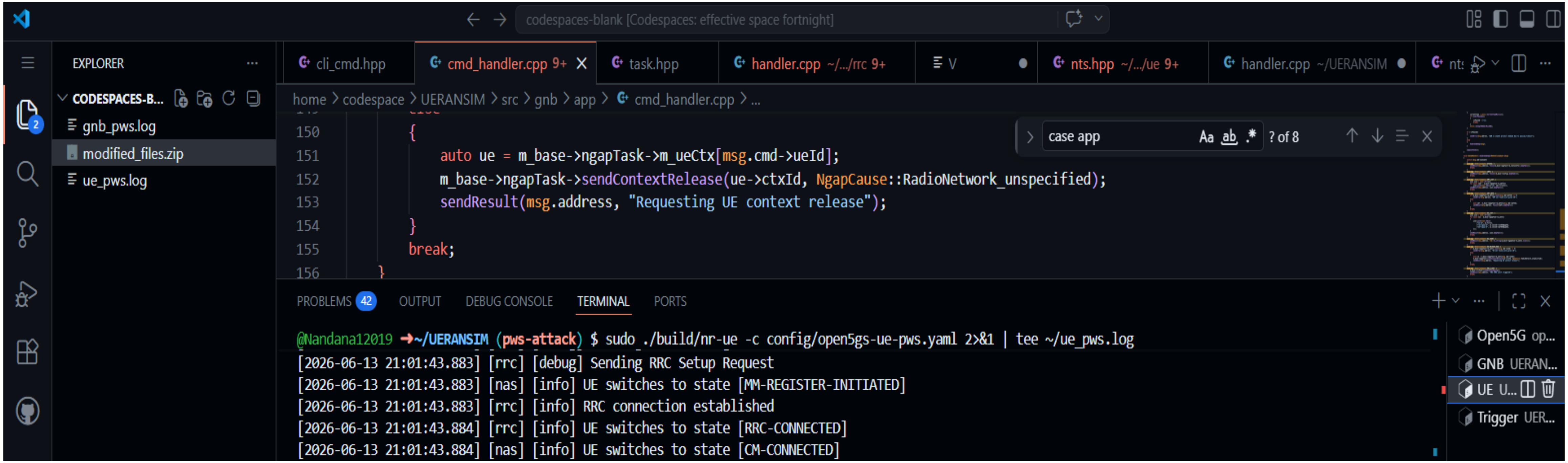
SIB1: si-BroadcastStatus = notBroadcasting
Paging: P-RNTI with ETWS indicator
PLMN: 999/70 (test network)

2. Victim UE State

Registered on fake network (RM-REGISTERED)
UE registers, transitions to CM-IDLE, remains camped on cell monitoring paging

3. Evidence Capture

Dual logs: gNB + UE perspective



Github Codespaces

RQ1 Validated: On-Demand PWS Forces Involuntary PRACH

gNB (UERANSIM)

[2026-05-31 12:02:14.784] [sctp] [info] Trying to establish SCTP connection... (127.0.0.5:38412)
[2026-05-31 12:02:14.790] [sctp] [info] SCTP connection established (127.0.0.5:38412)
[2026-05-31 12:02:14.791] [sctp] [debug] SCTP association setup ascId[240]
[2026-05-31 12:02:14.791] [ngap] [debug] Sending NG Setup Request
[2026-05-31 12:02:14.798] [ngap] [debug] NG Setup Response received
[2026-05-31 12:02:14.798] [ngap] [info] NG Setup procedure is successful

…

[2026-05-31 12:03:24.911] [rrc] [info] **[PWS-ATTACK] Broadcasting ETWS paging message** (P-RNTI broadcast)
[2026-05-31 12:03:24.912] [rrc] [info] **[PWS-ATTACK]** SIB1 configured with notBroadcasting for SIB6/7/8
[2026-05-31 12:03:24.912] [rrc] [info] **[PWS-ATTACK]** UEs in RRC_IDLE must request SIB6 via Random Access (PRACH)
[2026-05-31 12:03:24.912] [rrc] [info] **[PWS-ATTACK]** Sending broadcast paging -- all idle UEs will wake up
[2026-05-31 12:03:24.912] [rrc] [info] **[PWS-ATTACK]** Paging sent. UEs should now send RRCSystemInfoRequest via PRACH
[2026-05-31 12:03:24.923] [rrc] [info] **RRC Setup for UE[2]**
[2026-05-31 12:03:24.923] [rrc] [info] **RRC Setup for UE[3]**

UE (UERANSIM)

UERANSIM v3.2.8
[2026-05-31 12:02:17.845] [nas] [info] UE switches to state [MM-DEREGISTERED/PLMN-SEARCH]
[2026-05-31 12:02:17.846] [rrc] [debug] New signal detected for cell[1], total [1] cells in coverage
[2026-05-31 12:02:17.850] [rrc] [info] RRC connection established
[2026-05-31 12:02:17.850] [rrc] [info] UE switches to state [RRC-CONNECTED]
[2026-05-31 12:02:17.850] [nas] [info] UE switches to state [CM-CONNECTED]
[2026-05-31 12:02:17.855] [nas] [debug] Authentication Request received

…

[2026-05-31 12:03:04.543] [nas] [info] UE switches to state [CM-IDLE]
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK] *** ETWS PAGING RECEIVED — EMERGENCY ALERT AVAILABLE *****
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK]** Serving cell SIB1: si-BroadcastStatus = notBroadcasting for SIB6/7/8
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK]** Per 3GPP TS 38.331 §5.2.2.3.3: on-demand SI requires Random Access
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK]** *** INITIATING PRACH TO REQUEST SIB6 (ETWS PRIMARY ALERT) ***
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK]** **Sending RRC Setup Request — PRACH preamble transmitted on uplink**
[2026-05-31 12:03:24.922] [rrc] [info] **[PWS-ATTACK]** *** THIS UPLINK TRANSMISSION IS THE DETECTABLE SURVEILLANCE LEAK ***
[2026-05-31 12:03:24.922] [rrc] [debug] **Sending RRC Setup Request**
[2026-05-31 12:03:24.923] [rrc] [info] RRC connection established

gNB sends paging: 12:03:24.911

gNB detects UE: 12:03:24.923

**Validated on 3GPP-compliant simulator. Behaviour expected to be identical on commercial basebands (Qualcomm, Samsung Shannon, MediaTek) as the mandate is at the specification level (TS 38.331 §5.2.2.3.3), not implementation-specific.*

UE detects paging: 12:03:24.922

UE sends PRACH: 12:03:24.922

Result: 3GPP-compliant UE transmits PRACH when SIB delivery is on-demand

Presenter: Nandana Harikumar

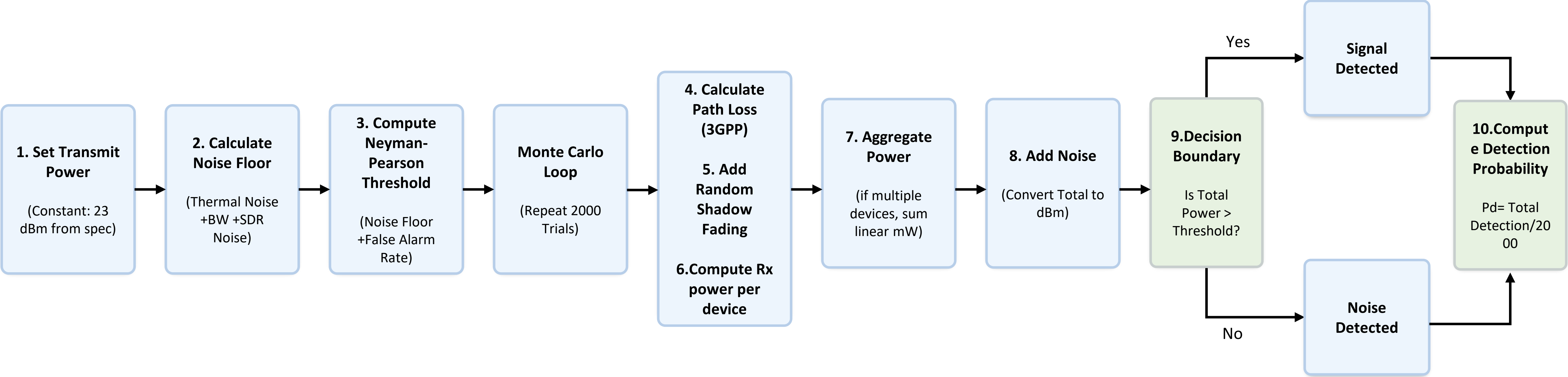
Sources: https://github.com/aligungr/UERANSIM , [4]

RQ2: PHYSICAL VALIDATION

RESEARCH QUESTION 2 - CAN PASSIVE EAVESDROPPING ON PRACH SIGNALS DETECT POPULATION PRESENCE?

Presenter: Nandana Harikumar

RQ2: Passive Detection of Mass PRACH Burst



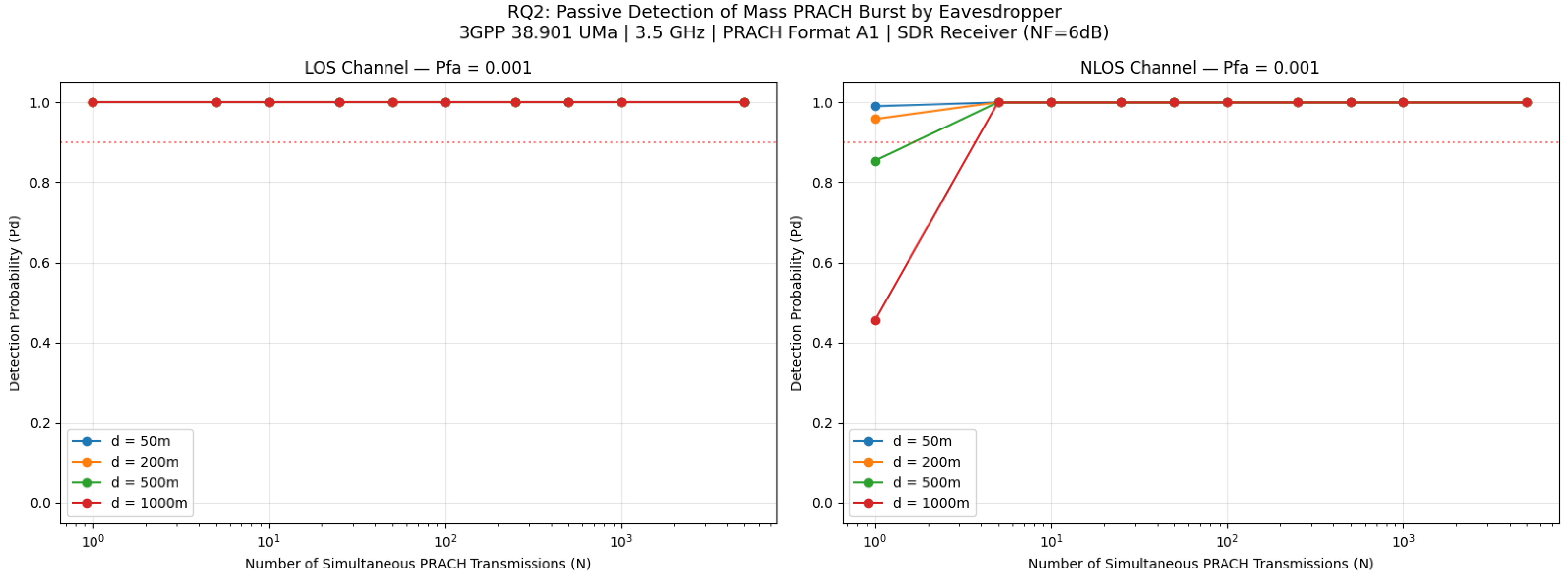
Simulation Setup	
1. Distances:	50m, 200m, 500m, 1000m
2. Devices (N):	1, 5, 10, 25, 50, 100, 500, 1000
3. Total Configurations:	4 distances × 9 device counts = 36 simulations
4. Each simulation:	2,000 Monte Carlo trials

Detection Model using 3GPP standardized parameters

Presenter: Nandana Harikumar

Sources: 3GPP TR 38.901 , 3GPP TS 38.211 , 3GPP TS 38.101-1

RQ2: Passive Detection of Mass PRACH Burst

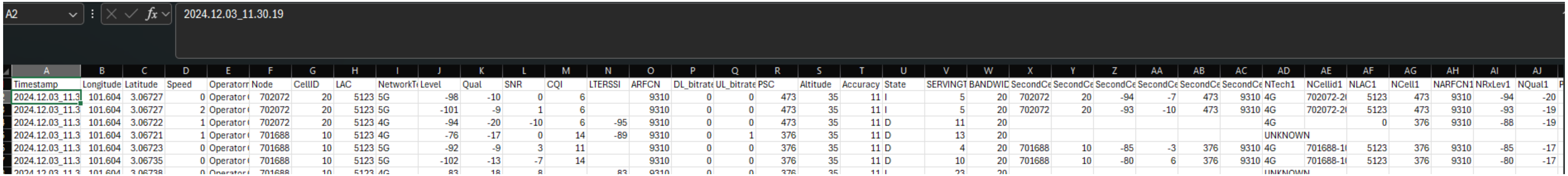


Result:

A passive eavesdropper with a $300 SDR achieves near-certain detection at 1km with just 6 simultaneously transmitting devices.

Presenter: Nandana Harikumar

Sunway University QoE-Aware Dataset



A2 | 2024.12.03_11.30.19

Timestamp	Longitude	Latitude	Speed	Operatorr	Node	CellID	LAC	NetworkT	Level	Qual	SNR	CQI
2024.12.03_11.3	101.604	3.06727	0	Operator (	702072	20	5123	5G	-98	-10	0	6
2024.12.03_11.3	101.604	3.06727	2	Operator (	702072	20	5123	5G	-101	-9	1	6
2024.12.03_11.3	101.604	3.06722	1	Operator (	702072	20	5123	4G	-94	-20	-10	6
2024.12.03_11.3	101.604	3.06721	1	Operator (	701688	10	5123	4G	-76	-17	0	14
2024.12.03_11.3	101.604	3.06723	0	Operator (	701688	10	5123	5G	-92	-9	3	11
2024.12.03_11.3	101.604	3.06735	0	Operator (	701688	10	5123	5G	-102	-13	-7	14

Timestamp	LTERSSI	ARFCN	DL_bitrate	UL_bitrate	PSC	Altitude	Accuracy	State	SERVINGT	BANDWID	SecondCe	SecondCe
2024.12.03_11.3		9310	0	0	473	35	11	I	5	20	702072	20
2024.12.03_11.3		9310	0	0	473	35	11	I	5	20	702072	20
2024.12.03_11.3	-95	9310	0	0	473	35	11	D	11	20		
2024.12.03_11.3	-89	9310	0	1	376	35	11	D	13	20		
2024.12.03_11.3		9310	0	0	376	35	11	D	4	20	701688	10
2024.12.03_11.3		9310	0	0	376	35	11	D	10	20	701688	10

Timestamp	SecondCe	SecondCe	SecondCe	SecondCe	NTech1	NCellid1	NLAC1	NCell1	NARFCN1	NRxLev1	NQual1
2024.12.03_11.3	-94	-7	473	9310	4G	702072-2(	5123	473	9310	-94	-20
2024.12.03_11.3	-93	-10	473	9310	4G	702072-2(	5123	473	9310	-93	-19
2024.12.03_11.3					4G		0	376	9310	-88	-19
2024.12.03_11.3					UNKNOWN						
2024.12.03_11.3	-85	-3	376	9310	4G	701688-1(	5123	376	9310	-85	-17
2024.12.03_11.3	-80	6	376	9310	4G	701688-1(	5123	376	9310	-80	-17

Device: Samsung Galaxy S21 Ultra with GNetTrack Pro

Environment: Dense urban, Node **701905**

Records: n = 6,016

Content: Extracted empirical path loss from RSRP at known UE-to-cell distances (Haversine).

Simulation: Re-ran Monte Carlo (2,000 trials) under both models with identical parameters.

Note: Path loss is reciprocal at 3.5 GHz TDD. Downlink RSRP measurements yield the same propagation channel as the uplink path.

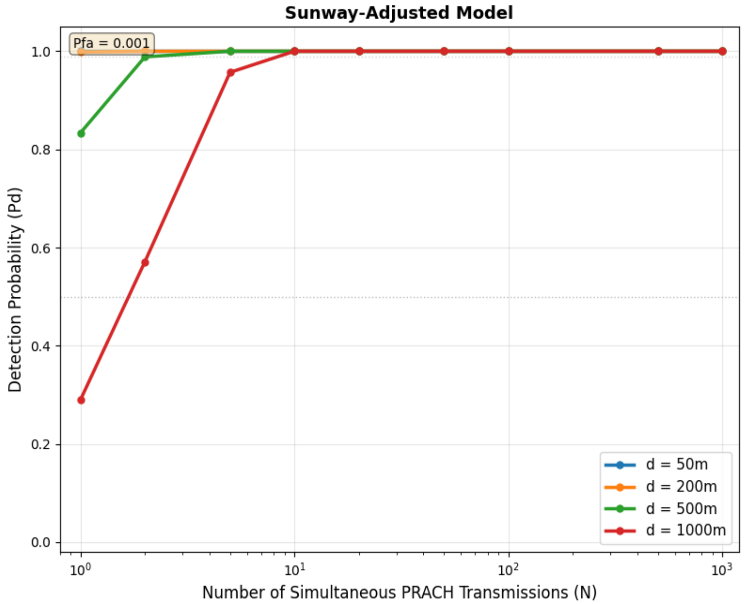
Dataset Statistics

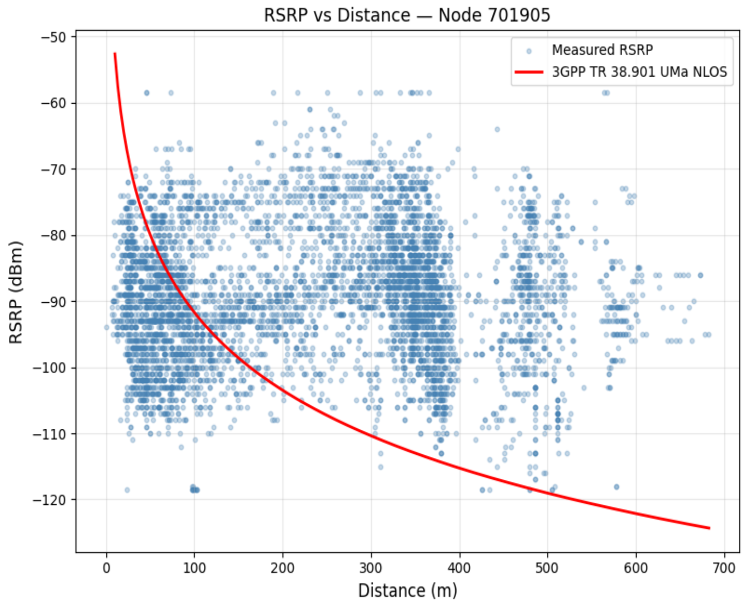
	Distance statistics (m)	RSRP statistics (dBm)
Count	6016.000000	6016.000000
Mean	238.835240	-89.377826
Std	158.504814	9.878238
Min	0.000000	-118.500000
25%	78.914039	-96.000000
50%	249.926268	-90.000000
75%	359.849988	-83.000000
max	682.631073	-58.500000
	Name: distance_m, dtype: float64	Name: Level, dtype: float64

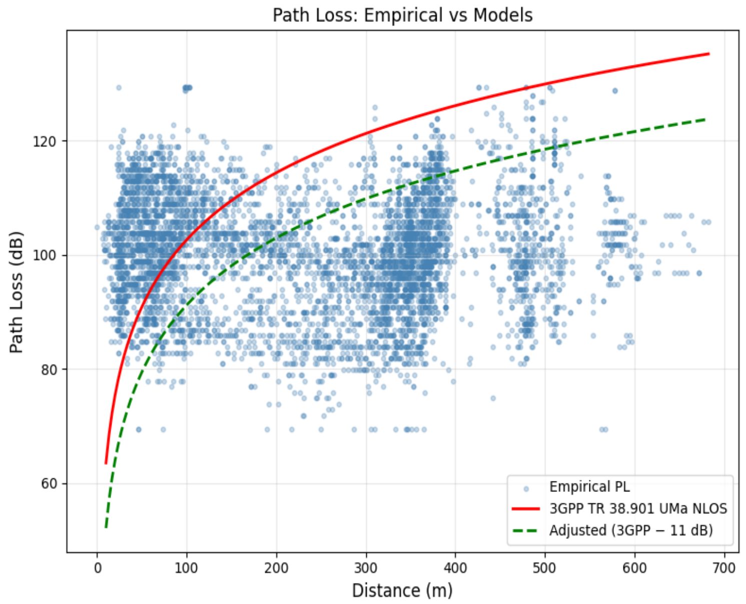
Presenter: Nandana Harikumar

Sources: https://data.mendeley.com/datasets/dx5xyyfz2y/1,[5]

Results







Distance bin (dist_bin)	Count	Mean Dist (mean_dist)	Mean RSRP (mean_rsrp)	Std RSRP (std_rsrp)	Mean PL (Empirical)	Mean PL (3GPP)	Mean Delta (mean_delta)	Std Delta (std_delta)
0-50m	768	33.19	-90.99	8.91	101.83	82.94	-18.89	9.78
50-150m	1556	86.2	-91.25	9.46	102.09	99.28	-2.81	10.65
150-300m	1064	226.53	-85.43	9.96	96.28	116.13	19.85	11.14
300-600m	2594	391.39	-89.39	9.95	100.24	125.48	25.24	9.76
600-1000m	33	634.05	-89.7	4.26	100.54	133.91	33.37	4.09

Verdict: Published results show single-device detection feasible at 500m and 10-device detection certain at 1km.

Presenter: Nandana Harikumar

Sources: 3GPP TR 38.901 , 3GPP TS 38.211 , 3GPP TS 38.101-1

Stated Limitations

RQ1 - Protocol Validation Limitations

- UERANSIM does not simulate the physical radio layer. Communication occurs over UDP loopback, not real RF. Physical-layer PRACH transmission is implicit - RRC Setup Request cannot occur without prior Random Access, but we did not observe an actual over-the-air preamble.
- Only one simulated UE was tested. A real attack targets hundreds of simultaneous devices. Scaling behavior (PRACH collisions, timing spread across DRX cycle) was not validated in the testbed.

RQ2 - Detection Simulation Limitations

- Sunway dataset covers 0–700m. The 1km detection results extrapolate beyond the measured range using the fitted constant offset. The actual delta at 600-1000m is ~33 dB (higher than the mean 11 dB), meaning extrapolation is conservative.
- The constant offset model (PL_adj = PL_3GPP - 11 dB) is a simplification. The empirical delta varies from -19 dB at close range to +33 dB at long range. The mean offset underestimates the correction at operationally relevant distances (500m-1km).
- Shadow fading is modelled as independent across devices. In reality, devices in close proximity may experience correlated shadowing. This could cause the aggregate energy to be more variable than the simulation predicts.
- The simulation places all devices at a fixed distance from the eavesdropper (Sunway comparison) or uniformly distributed in a disk (original simulation). Real device distributions may be clustered (e.g., inside a building) which would change the energy profile.

Threat Model Limitations

- Only a fraction of UEs will camp on the rogue cell. Depends on relative signal strength, cell reselection parameters, and UE state. Even partial capture is useful - triggering 20% of devices and detecting N PRACH bursts estimates total population as ~5N.

Presenter: Nandana Harikumar

Sources: 3GPP TR 38.901 , 3GPP TS 38.211 , 3GPP TS 38.101-1

Countermeasures

SIB1 Authentication

Digital signatures on SIB1 broadcast. UE verifies signature before trusting si-BroadcastStatus field.

PRACH Timing Randomization

UE adds a random delay (0–500ms) before sending SI request via PRACH. Spreads the synchronized burst across time, making aggregate energy detection harder.

SIB Broadcast Mandate for PWS

3GPP mandates that SIB6/7/8 must always use broadcast mode. notBroadcasting disallowed for PWS SIBs.

Fake Base Station Detection Heuristics

UE cross-checks cell parameters (PLMN, TAC, signal strength, SIB configuration) against known legitimate cells or operator-provided whitelists.

Conclusion: Spoofed Public Warning System Alerts as a Bulk Surveillance Primitive

RQ1 - Mandatory Uplink Leakage

Does 3GPP PWS guarantee an involuntary uplink transmission from all proximate UEs?

[i] 3GPP TS 38.331 §5.2.2.3.3 mandates PRACH on SIB6/7/8 reception - no UE opt-out mechanism exists within the standard

[ii] UEs cannot authenticate SIB1 or verify cell legitimacy; protocol compliance is unconditional

[iii] Validated via modified UERANSIM: ETWS paging → CM-IDLE → PRACH → RRC-CONNECTED across two independent log sources at matching ms timestamps

RQ2 - Zero-Knowledge Presence Detection

Can an eavesdropper verify mass device presence without recovering any UE identity?

[i] Neyman-Pearson detector at Pfa = 0.001 achieves 99.5% detection for N = 5 devices at 1 km urban NLOS using commodity SDR (~$300 USD)

[ii] No IMSI, SUCI, or cryptographic decryption required - energy-level observation alone is sufficient for confirmation

[iii] Sunway QoE-Aware Dataset (Samsung S21 Ultra) confirms simulation is conservative: real urban signals are 58 dB stronger than modelled

Overall Verdict:

Hypothesis confirmed across two independent research questions - theoretical analysis corroborated by experimental validation.

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Disclosure: Content was enhanced with AI-assisted editing for technical clarity and flow. Implementation (Gemini 3.1 pro, Claude 4.6 via VS Code)